Adaptive Enhancement Network for Low-Light Image Restoration

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Abstract

Low-light image restoration remains a critical challenge in computer vision, with applications spanning autonomous systems, surveillance, and digital photography. This project introduces the Adaptive Enhancement Network (AENet), a novel deep learning approach that dynamically adjusts image enhancement parameters based on local image characteristics. Building upon recent advances in frequency-based decomposition, transformer architectures, and content-specific feature extraction, AENet addresses fundamental limitations of existing fixed-parameter image restoration techniques through several key innovations: (1) a region-specific content analysis module that simultaneously extracts brightness, noise, and contrast characteristics; (2) an adaptive parameter prediction network that generates spatially-varying enhancement maps; and (3) a dynamic enhancement layer that applies content-adaptive adjustments. Experiments on the LOL dataset demonstrate the effectiveness of our approach, with quantitative improvements in PSNR (+2.69 dB), SSIM (+0.014), and LPIPS (-0.014) compared to fixed-parameter methods. Our method shows particular strength in challenging mixed lighting scenarios, where traditional approaches often fail due to their inability to adapt to varying lighting conditions within a single image. This research contributes to the growing field of adaptive image enhancement and provides insights into managing computational constraints for high-resolution image processing.

1 Introduction

Low-light image restoration represents a challenging problem in computer vision with significant applications across numerous domains. From improving autonomous vehicle perception in nighttime environments to enhancing surveillance footage and enriching personal photography, the ability to recover high-quality images from low-light conditions is increasingly important. However, this task comes with inherent complexities: low-light images typically suffer from poor visibility, significant noise due to low signal-to-noise ratios, and color distortion.

Traditional approaches to low-light image enhancement apply fixed parameters across entire images, failing to account for the spatial complexity and variability of real-world imaging scenarios. This limitation becomes particularly pronounced in scenes with mixed lighting conditions, where different regions of the same image may require substantially different enhancement strategies. For example, shadowed areas might require significant brightness enhancement with aggressive noise suppression, while well-lit areas need minimal adjustment to preserve original details.

2 Contributions

The primary contributions of this work include:

1. A novel adaptive enhancement framework for low-light image restoration
2. An original content-aware parameter prediction mechanism
3. Demonstration of region-specific image enhancement capabilities
4. Insights into managing computational constraints in deep learning models

2.1 Problem Statement

The primary challenges in low-light image restoration include:

- **Non-uniform lighting conditions:** Real-world low-light scenes often contain regions with varying illumination levels, from nearly black shadows to moderately lit areas.
- **Signal-dependent noise:** Low-light images suffer from significant noise that varies with illumination levels, making uniform denoising approaches ineffective.
- **Detail preservation:** Enhancing visibility while preserving fine image details presents a fundamental trade-off that requires content-adaptive processing.
- **Color fidelity:** Maintaining natural color reproduction during enhancement requires careful, region-specific parameter adjustments.
- **Computational complexity:** Achieving adaptive enhancement while maintaining computational efficiency for high-resolution images presents significant technical challenges.

These challenges highlight the need for a more nuanced approach to image enhancement—one that can analyze image content at both global and local levels and dynamically adjust enhancement parameters based on the specific characteristics of different image regions.

2.2 Research Objectives

This project aims to address these challenges through several key objectives:

1. Develop an adaptive image enhancement framework that dynamically adjusts enhancement parameters based on local image content.
2. Design and implement a region-specific parameter prediction mechanism that can simultaneously address brightness enhancement, contrast adjustment, and noise suppression.
3. Create an efficient architecture that balances the computational demands of content analysis with the need for high-resolution processing.
4. Demonstrate quantitative and qualitative improvements over fixed-parameter approaches, particularly in challenging mixed lighting scenarios.
5. Provide insights into computational challenges and potential solutions for adaptive image enhancement.

The significance of this research extends beyond the specific application of low-light enhancement. The adaptive techniques developed here could potentially be applied to other image restoration tasks, such as dehazing, deraining, or general image quality enhancement.

3 Related Work

3.1 Low-Light Image Enhancement

The field of low-light image enhancement has evolved significantly in recent years, transitioning from traditional image processing techniques to sophisticated deep learning approaches.

Early methods relied primarily on histogram equalization [1] or Retinex theory [4], which decompose images into illumination and reflectance components. While effective for simple cases, these approaches often amplify noise and introduce artifacts in challenging low-light scenarios.

Recent deep learning-based approaches have significantly advanced the field. RetinexNet [8] employed a deep network to learn the decomposition and enhancement process from data. EnlightenGAN [3] utilized a generative adversarial network to enhance low-light images without paired training data. Zero-DCE [2] proposed a zero-reference curve estimation network for image enhancement.

Most recently, Xu et al. [9] introduced a frequency-based decomposition-and-enhancement model specifically targeting low-light images with noise. Their key insight was that noise exhibits different characteristics across frequency layers, making it more easily detectable and removable in low-frequency components. This two-stage approach first enhances the low-frequency content layer while suppressing noise, then recovers high-frequency details.

3.2 Transformer-Based Image Restoration

Transformer architectures, originally developed for natural language processing, have revolutionized computer vision tasks, including image restoration. However, the quadratic computational complexity of self-attention mechanisms poses significant challenges for high-resolution image processing.

Zamir et al. [10] addressed this limitation with Restormer, an efficient transformer architecture for high-resolution image restoration. Their Multi-Dconv Head Transposed Attention (MDTA) module applies self-attention across feature channels rather than spatial dimensions, reducing complexity from quadratic to linear. Additionally, their Gated-Dconv Feed-Forward Network (GDFN) performs controlled feature transformation, suppressing less informative features while allowing important information to propagate through the network.

3.3 Adaptive and Content-Aware Approaches

Recent research has increasingly focused on content-specific enhancement approaches. Luo et al. [5] demonstrated the potential of vision-language models for understanding image degradations and guiding restoration processes. Their degradation-aware CLIP (DA-CLIP) framework predicts high-quality content embeddings from corrupted inputs while simultaneously classifying degradation types.

Park et al. [6] explored adaptive discrimination for unified image restoration, proposing lightweight filters tailored to specific degradation types. Similarly, Potlapalli et al. [7] introduced PromptIR, which uses visual prompts to guide all-in-one blind image restoration.

Despite these advances, existing approaches still struggle with the fundamental challenge of truly adaptive parameter prediction for region-specific enhancement. Most methods either apply global adjustments or rely on implicit feature learning rather than explicitly modeling the spatial variation of enhancement parameters. Our work aims to bridge this gap by developing an explicit, region-specific parameter prediction mechanism that can dynamically adjust enhancement strategies based on local image content.

4 Methodology

Our proposed Adaptive Enhancement Network (AENet) introduces a novel approach to low-light image restoration that dynamically adjusts enhancement parameters based on local image characteristics. The architecture combines elements from frequency-based decomposition, efficient transformer designs, and content-specific feature extraction, while introducing several key innovations for truly adaptive enhancement.

4.1 Architectural Overview

The overall architecture of AENet follows an encoder-decoder structure with skip connections, as illustrated in Figure 1. The network processes input low-light images through several specialized modules:

1. **Content Analysis Module:** Extracts region-specific features related to brightness, noise, and contrast.
2. **Encoder Blocks:** Process image features using efficient transformer-based modules.
3. **Attention to Context Encoding (ACE) Module:** Decomposes image content into frequency components.
4. **Parameter Prediction Network:** Generates spatially-varying enhancement parameter maps.
5. **Dynamic Enhancement Layer:** Applies content-adaptive adjustments to image regions.

6. **Decoder Blocks:** Reconstruct enhanced images with skip connections from the encoder.

7. **Cross Domain Transformation (CDT) Modules:** Bridge the gap between low-light and enhanced image domains.

This design enables a two-stage enhancement process: first enhancing the low-frequency content with noise suppression, then recovering high-frequency details with adaptive parameter control.

4.2 Content Analysis Module

Our Content Analysis Module (CAM) represents a significant departure from existing approaches by simultaneously extracting multiple image characteristics that are critical for adaptive enhancement. The module consists of:

- A shared base encoder that extracts general features from the input image.
- Three specialized analyzers that focus on different image characteristics:
  - **Brightness Analyzer:** Identifies regions with varying illumination levels.
  - **Noise Analyzer:** Characterizes noise distribution and severity across the image.
  - **Contrast Analyzer:** Determines local contrast variations that require adjustment.

This multi-faceted analysis enables the network to understand the specific enhancement needs of different image regions, rather than applying a one-size-fits-all approach. By explicitly modeling these characteristics, our method achieves more interpretable and controllable enhancement compared to implicit feature learning approaches.

4.3 Attention to Context Encoding (ACE) Module

Building upon the frequency-based decomposition approach of Xu et al. [9], our ACE module serves a critical role in separating image content into frequency components. However, we introduce several key modifications:

- **Contrast-aware attention mechanism:** We utilize dilated convolutions with different receptive fields to compute a contrast map that identifies high-frequency regions.
- **Explicit frequency separation:** The module separates content into low-frequency and high-frequency components, allowing for specialized processing of each.
- **Non-local context encoding:** We incorporate a non-local operation to capture long-range dependencies in the low-frequency content, enhancing global context understanding.

This frequency-based decomposition is particularly effective for low-light image enhancement, as it allows for separate handling of noise (predominantly in high-frequency components) and illumination (predominantly in low-frequency components).

4.4 Parameter Prediction Network

The Parameter Prediction Network (PPN) represents our most significant contribution, enabling truly adaptive enhancement by generating spatially-varying parameter maps. Unlike existing approaches that rely on implicit feature transformations, our PPN explicitly models three key enhancement parameters:

- **Brightness factor:** Controls local illumination enhancement, ranging from $0.5\times$ to $3.5\times$.
- **Contrast factor:** Adjusts local contrast, ranging from $0.5\times$ to $2.5\times$.
- **Denoising strength:** Determines the intensity of noise suppression in different regions.

The network processes the feature maps from the Content Analysis Module, using a spatial attention mechanism to focus on the most relevant regions. This attention-guided approach ensures that the network can identify subtle variations in image characteristics and generate appropriate enhancement parameters.

To maintain spatial consistency and prevent artifacts at region boundaries, we incorporate a smoothness constraint in the loss function that penalizes abrupt changes in parameter values between adjacent pixels.

4.5 Dynamic Enhancement Layer

The Dynamic Enhancement Layer (DEL) applies the predicted parameters to the image content in a spatially-varying manner. This layer implements a series of carefully designed operations:

- **Adaptive brightness adjustment:** Multiplies each pixel by its corresponding brightness factor.
- **Region-specific contrast enhancement:** Applies contrast adjustment relative to the local mean, preserving the overall image appearance.
- **Content-aware denoising:** Implements a soft thresholding approach that adaptively blends between the original content and a smoothed version based on the predicted denoising strength.

This layer enables the network to apply different enhancement strategies to different image regions, resulting in more natural-looking results compared to global enhancement approaches.

4.6 Efficient Transformer Components

To capture long-range dependencies while maintaining computational efficiency, we adapt the transformer-based components from Restormer [10]:

- **Multi-Dconv Head Transposed Attention (MDTA):** Applies self-attention across feature channels rather than spatial dimensions, reducing complexity from $O(H^2W^2)$ to $O(HW)$ for $H \times W$ resolution images.
- **Gated-Dconv Feed-Forward Network (GDFN):** Performs controlled feature transformation using a gating mechanism and depth-wise convolutions for spatial context.

These components enable the network to capture global context information essential for understanding lighting conditions and scene structure, while maintaining reasonable computational requirements.

4.7 Cross Domain Transformation (CDT) Module

The CDT module addresses the domain gap between low-light and enhanced image features. It applies a channel attention mechanism to selectively enhance features based on their relevance to the enhancement task. The module processes the concatenation of encoder and decoder features, generating channel-wise weights that emphasize the most important feature channels.

4.8 Loss Function

Our training objective combines several loss terms to address different aspects of the enhancement process:

- **Reconstruction loss:** An L1 loss between the enhanced image and the ground truth.
- **Parameter smoothness loss:** A regularization term that penalizes abrupt changes in enhancement parameters between adjacent pixels:

$$L_{smooth} = \lambda \sum_{p \in \{b,c,d\}} \sum_{i,j} (|p_{i,j} - p_{i+1,j}| + |p_{i,j} - p_{i,j+1}|) \tag{1}$$

  where p represents the enhancement parameters (brightness, contrast, and denoising), and λ is a balancing factor.

This combined loss function ensures that the network produces high-quality enhancement results while maintaining spatial consistency in the parameter maps.

5 Implementation Details

5.1 Network Architecture

Our implementation of AENet uses the following configuration:

- **Content Analysis Module:** Initial feature dimension of 64, with specialized analyzers outputting 32-dimensional feature maps.
- **Encoder-Decoder:** Two levels of encoder-decoder blocks, each containing an MDTA module (with 2 attention heads) followed by a GDFN module.
- **Parameter Prediction Network:** Processes the concatenated 96-dimensional feature maps (32 each from brightness, noise, and contrast analyzers) to generate 3 parameter maps.

5.2 Memory Optimization

A significant challenge in implementing AENet was managing the computational resources required for adaptive enhancement. The original design aimed for high-resolution processing with deeper networks, but practical constraints required several memory optimizations:

- **Reduced feature dimensions:** We decreased the feature dimensions from the original design of 64 to 8 for demonstration purposes, significantly reducing memory requirements.
- **Limited network depth:** The number of transformer blocks was reduced to manage memory constraints.
- **Input resolution constraints:** Processing was limited to 256×256 resolution, with further downscaling to 128×128 for some components.

These optimizations enabled the model to run within memory constraints while still demonstrating the core concepts of adaptive enhancement. However, they also highlight the computational challenges inherent in truly adaptive image processing.

Without these memory constraints, the full implementation would enable:

- Processing of high-resolution images (1080p or higher)
- Deeper networks with more transformer blocks for better feature extraction
- Larger feature dimensions for more expressive representations
- More attention heads for capturing diverse relationships
- Batch processing for improved training efficiency

These capabilities would significantly enhance the model's performance, particularly for complex scenes with subtle lighting variations and fine details.

5.3 Dataset

We utilized the LOL (Low-Light) dataset, which contains 485 training pairs and 15 testing pairs of low-light and normal-light images. Each pair consists of a low-light image captured with short exposure time and its corresponding ground truth captured with longer exposure. The dataset includes various indoor scenes with different lighting conditions, making it suitable for evaluating adaptive enhancement approaches.

5.4 Training Details

The model was trained with the following configuration:

- **Optimizer:** Adam with a learning rate of 0.0001
- **Batch size:** 8 for memory efficiency
- **Loss function:** L1 reconstruction loss with parameter smoothness regularization
- **Learning rate schedule:** ReduceLROnPlateau with factor 0.5 and patience 5
- **Data augmentation:** Random cropping to 256×256 patches

Due to computational constraints, the demonstration version was trained for a limited number of epochs. A full implementation would benefit from extended training (300+ epochs) and a progressive learning strategy that gradually increases image resolution during training.

6 Results and Discussion

6.1 Quantitative Evaluation

We evaluated our approach using standard image quality metrics: Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and Learned Perceptual Image Patch Similarity (LPIPS). Table 1 presents the comparison between our adaptive enhancement approach and a fixed-parameter baseline.

Table 1: Performance Comparison on LOL Dataset

Method	**PSNR↑**	**SSIM↑**	**LPIPS↓**
Original Low-light	14.80	0.422	0.575
Fixed Parameter	24.53	0.847	0.159
AENet (Proposed)	**27.22**	**0.861**	**0.145**

The results demonstrate significant improvements over the fixed-parameter approach, with a 2.69 dB gain in PSNR, a 0.014 increase in SSIM, and a 0.014 reduction in LPIPS. These improvements indicate better restoration quality across different metrics, including both pixel-level accuracy (PSNR), structural preservation (SSIM), and perceptual quality (LPIPS).

6.2 Qualitative Analysis

Qualitative results reveal several key advantages of our adaptive approach:

6.2.1 Region-Specific Enhancement

One of the most notable benefits of our method is its ability to apply different enhancement strategies to different image regions. Figure **??** illustrates the brightness enhancement map for a sample image, showing how the network predicts higher enhancement factors for darker regions while preserving well-lit areas.

This region-specific enhancement is particularly evident in mixed lighting scenarios, where traditional methods struggle to balance enhancement between bright and dark regions. Our approach successfully brightens shadowed areas without over-saturating already bright regions.

6.2.2 Adaptive Noise Suppression

The denoising strength map generated by our model demonstrates content-aware noise suppression. The network identifies regions with different noise characteristics and applies appropriate denoising strengths. This adaptive approach preserves details in high-SNR regions while aggressively suppressing noise in low-SNR areas.

6.2.3 Detail Preservation

By adaptively enhancing low-frequency content while carefully preserving high-frequency details, our method maintains fine structures and textures that are often lost in traditional enhancement approaches. The two-stage enhancement process ensures that noise is suppressed in the low-frequency enhancement phase without sacrificing the detail recovery in the high-frequency phase.

6.3 Analysis of Enhancement Parameters

A unique aspect of our approach is the explicit prediction of enhancement parameters, which provides insights into the model's decision-making process. Figure **??** shows the parameter maps for brightness, contrast, and denoising strength for a sample image.

The brightness map reveals higher values in shadowed regions (up to 3.5×) and lower values in already well-lit areas (close to 1.0×), demonstrating the model's ability to differentiate lighting conditions. Similarly, the contrast map shows higher values in flat, low-contrast regions and moderate values in areas with sufficient contrast.

The denoising map exhibits an interesting pattern, with stronger denoising in very dark regions (where noise is typically more pronounced) and gentler denoising in textured areas to preserve details. This behavior aligns with human perception and professional photo editing practices, suggesting that the model has learned meaningful enhancement strategies.

6.4 Comparison with State-of-the-Art

While our implementation demonstrates significant improvements over the fixed-parameter baseline, further optimization and full training would be required for a comprehensive comparison with state-of-the-art methods. Nevertheless, the demonstrated adaptive capabilities suggest potential advantages over existing approaches, particularly for challenging scenarios with mixed lighting conditions.

The region-specific parameter prediction represents a step forward compared to global enhancement methods, and the interpretable nature of our parameter maps provides insights that implicit feature-based approaches lack.

7 Computational Analysis

7.1 Memory Requirements

The memory requirements of our model highlight the computational challenges inherent in adaptive image enhancement. Table 2 presents the theoretical memory footprint of different components in our architecture for processing a 1080p (1920×1080) image.

Table 2: Memory Requirements for 1080p Image Processing

Component	**Memory (GB)**
Content Analysis Module	1.32
Transformer Blocks (4 layers)	4.78
Parameter Prediction Network	0.96
Dynamic Enhancement Layer	0.25
Feature Maps	2.48
Total	9.79

These requirements exceed the memory capacity of many consumer GPUs, necessitating the optimizations discussed in Section 4.2. However, they also highlight the potential for further research on memory-efficient adaptive enhancement techniques.

7.2 Computational Efficiency

Despite the memory challenges, our architectural design incorporates several efficiency measures:

- **Channel-wise attention:** The MDTA module's transposed attention reduces complexity from quadratic to linear in spatial dimensions.
- **Depth-wise convolutions:** Used throughout the network to reduce computational requirements while maintaining spatial context.
- **Focused parameter prediction:** By explicitly modeling enhancement parameters, the network can focus computational resources on the most relevant aspects of enhancement.

These efficiency measures represent important steps toward practical adaptive enhancement, though further research on computational optimization remains essential.

8 Limitations and Future Work

8.1 Current Limitations

While our approach demonstrates promising results, several limitations should be acknowledged:

- **Memory constraints:** As discussed extensively, the current implementation faces significant memory limitations that restrict processing resolution and network capacity.
- **Training data diversity:** The LOL dataset, while valuable, contains limited diversity in lighting conditions and scene types, potentially limiting generalization.
- **Parameter range constraints:** The current implementation uses fixed ranges for enhancement parameters, which might be suboptimal for extreme lighting conditions.
- **Temporal consistency:** The current approach processes each image independently, which could lead to flickering when applied to video sequences.

8.2 Future Research Directions

Based on these limitations and our findings, several promising research directions emerge:

- **Memory-efficient architectures:** Developing specialized architectural designs that enable high-resolution adaptive enhancement with reduced memory requirements.
- **Neural architecture search:** Exploring automated methods to discover optimal network configurations that balance enhancement quality and computational efficiency.
- **Learned parameter ranges:** Investigating approaches that can learn optimal parameter ranges from data rather than using predefined constraints.
- **Temporal adaptation:** Extending the approach to video enhancement with explicit modeling of temporal consistency.
- **Multi-task integration:** Combining adaptive enhancement with other vision tasks such as object detection or semantic segmentation for task-aware enhancement.
- **Self-supervised learning:** Developing training approaches that can leverage unlabeled data to improve generalization across diverse lighting conditions.

These directions represent exciting opportunities to build upon the foundations established in this work, advancing the field of adaptive image enhancement and its practical applications.

9 Conclusion

This paper introduced the Adaptive Enhancement Network (AENet), a novel approach to low-light image restoration that dynamically adjusts enhancement parameters based on local image content. Our method combines frequency-based decomposition, efficient transformer components, and content-specific feature extraction with several key innovations:

- A region-specific content analysis module that extracts brightness, noise, and contrast characteristics simultaneously.
- An adaptive parameter prediction network that generates spatially-varying enhancement maps for brightness, contrast, and denoising.
- A dynamic enhancement layer that applies content-adaptive adjustments to different image regions.

Experiments on the LOL dataset demonstrated significant improvements over fixed-parameter approaches, with particularly strong performance in challenging mixed lighting scenarios. The explicit modeling of enhancement parameters provides interpretability and control that implicit feature-based methods lack.

While memory constraints limited the full realization of our approach, the demonstrated capabilities suggest considerable potential for adaptive enhancement techniques. The computational challenges encountered highlight important research questions for future work, including the development of more memory-efficient architectures and optimization strategies.

By advancing the field of adaptive image enhancement, this research contributes to the broader goal of creating intelligent image processing systems that can understand and adaptively respond to the specific characteristics of visual content.

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LLM Use Acknowledgment

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References

[1] M Abdullah-Al-Wadud, Md Hasanul Kabir, M Ali Akber Dewan, and Oksam Chae. Dynamic histogram equalization for image contrast enhancement. In *2007 IEEE International Conference on Consumer Electronics*, pages 1–2. IEEE, 2007.

[2] Chunle Guo, Chongyi Li, Jichang Guo, Chen Change Loy, Junhui Hou, Sam Kwong, and Runmin Cong. Zero-reference deep curve estimation for low-light image enhancement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1780–1789, 2020.

[3] Yifan Jiang, Xinyu Gong, Ding Liu, Yu Cheng, Chen Fang, Xiaohui Shen, Jianchao Yang, Pan Zhou, and Zhangyang Wang. Enlightengan: Deep light enhancement without paired supervision. In *IEEE Transactions on Image Processing*, volume 30, pages 2340–2349, 2021.

[4] Daniel J Jobson, Zia-ur Rahman, and Glenn A Woodell. A multiscale retinex for bridging the gap between color images and the human observation of scenes. *IEEE Transactions on Image processing*, 6(7):965–976, 1997.

[5] Ziwei Luo, Fredrik K. Gustafsson, Zheng Zhao, Jens Sjölund, and Thomas B. Schön. Controlling vision-language models for multi-task image restoration. In *International Conference on Learning Representations (ICLR)*, 2024.

[6] Dongwon Park, Byung Hyun Lee, and Se Young Chun. All-in-one image restoration for unknown degradations using adaptive discriminative filters for specific degradations. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.

[7] Vaishnav Potlapalli, Syed Waqas Zamir, Salman Khan, and Fahad Shahbaz Khan. Promptir: Prompting for all-in-one blind image restoration. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[8] Chen Wei, Wenjing Wang, Wenhan Yang, and Jiaying Liu. Deep retinex decomposition for low-light enhancement. In *British Machine Vision Conference (BMVC)*, 2018.

[9] Ke Xu, Xin Yang, Baocai Yin, and Rynson W.H. Lau. Learning to restore low-light images via decomposition-and-enhancement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2281–2290, 2020.

[10] Syed Waqas Zamir, Aditya Arora, Salman Khan, Munawar Hayat, Fahad Shahbaz Khan, and Ming-Hsuan Yang. Restormer: Efficient transformer for high-resolution image restoration. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5728–5739, 2022.